

Microencapsulation as a Strategy for Precision Microbiome Engineering in Superfund Site Remediation

Duke Superfund Research Center

Project 5 – Precision Bioremediation

Erica Babusci – Gunsch Lab

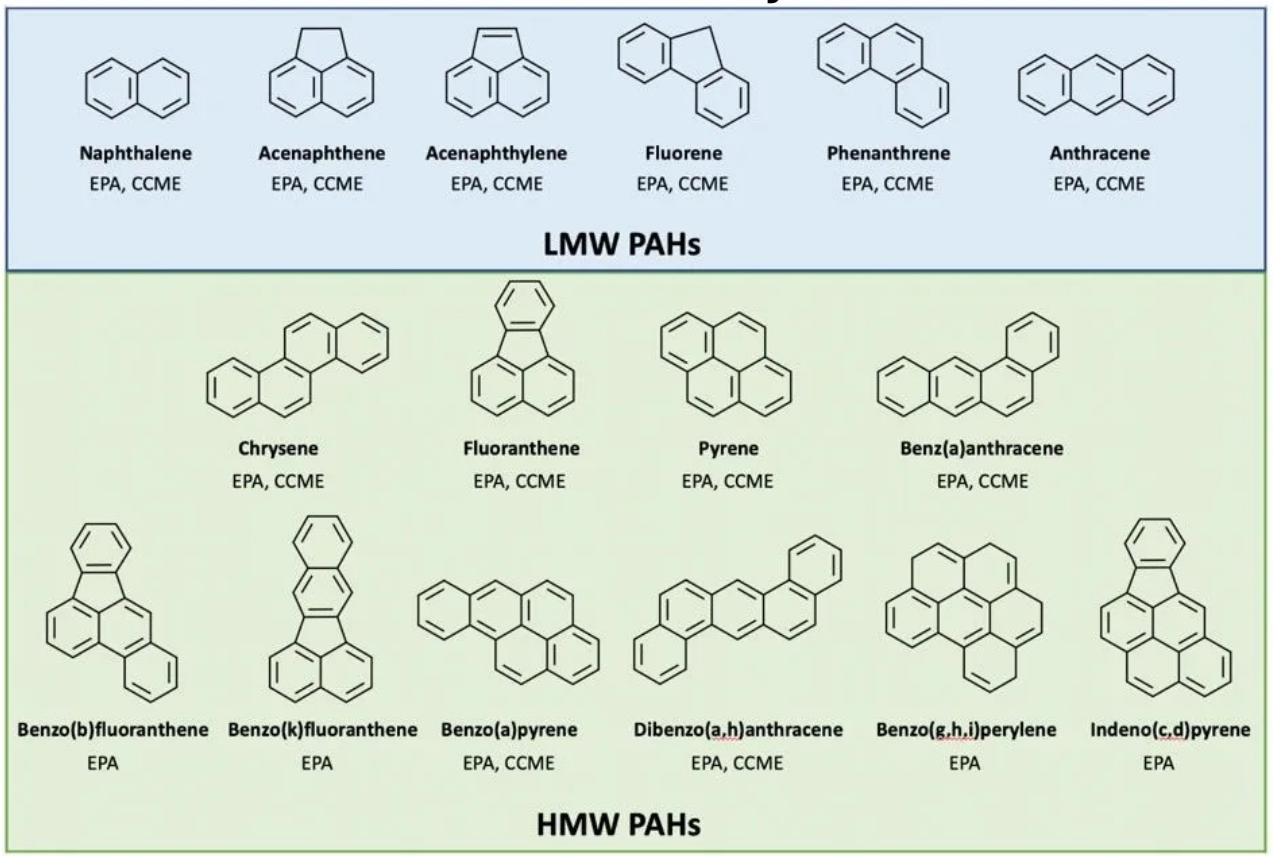
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Early Life Exposures | Later Life Consequences

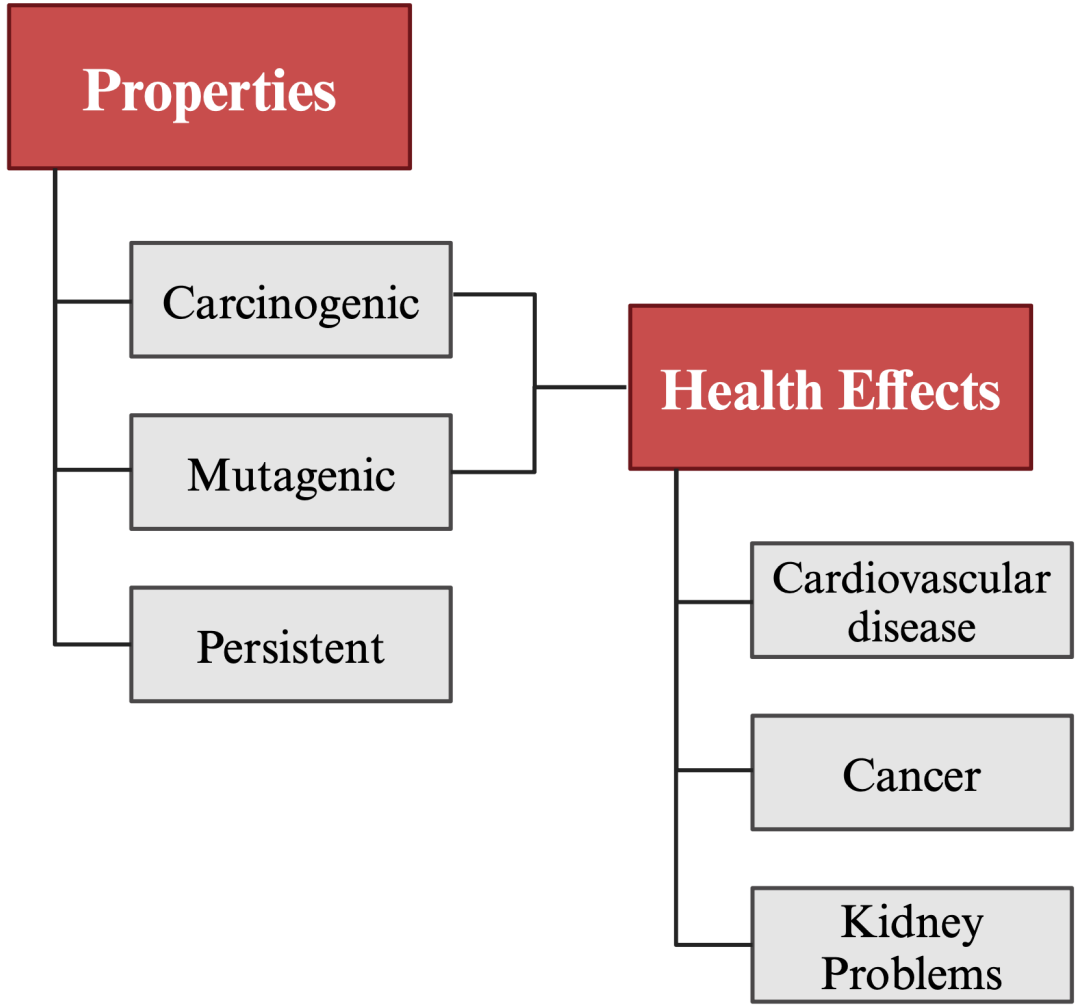
NIH NIEHSSRP

Polycyclic aromatic hydrocarbons (PAHs)

16 PAHs on the EPA's Priority Pollutant List



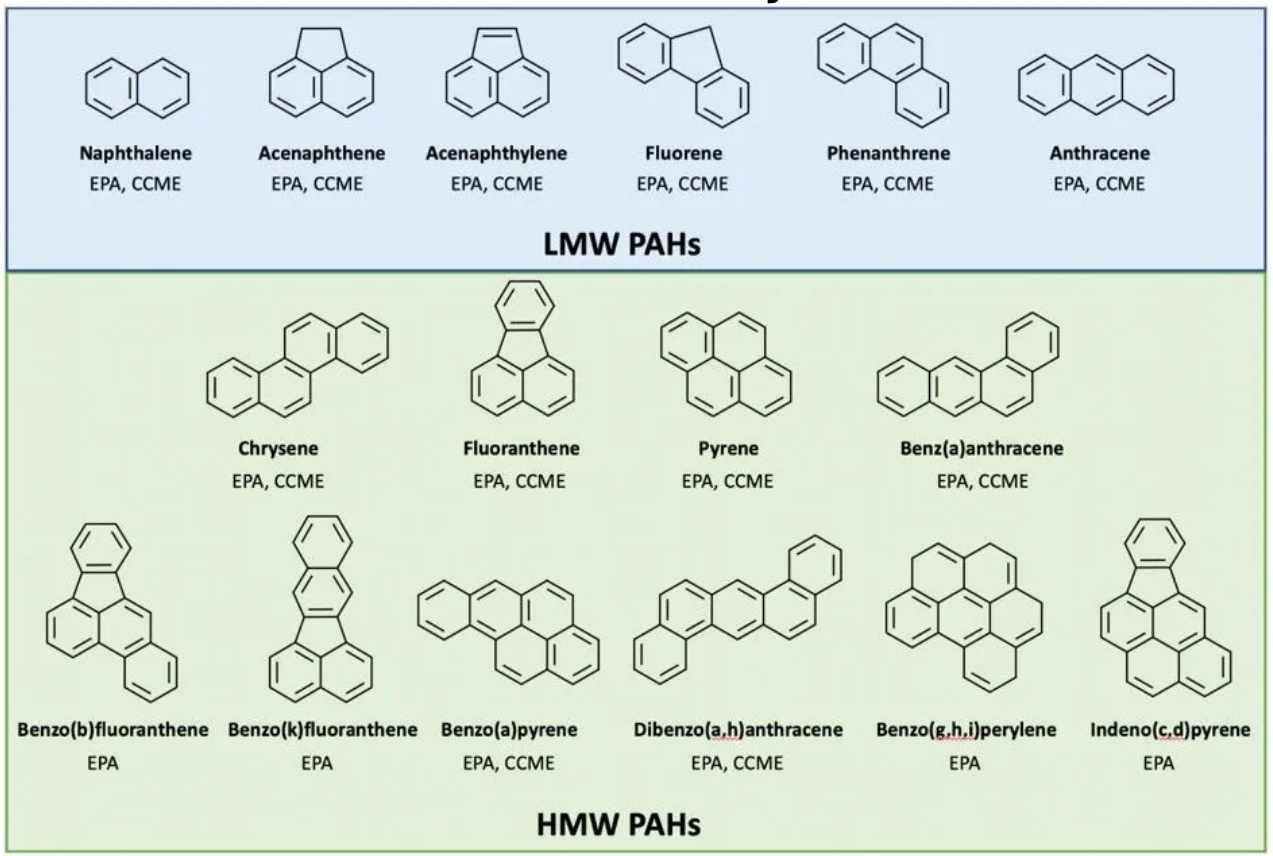
Chemistry Matters, 2021



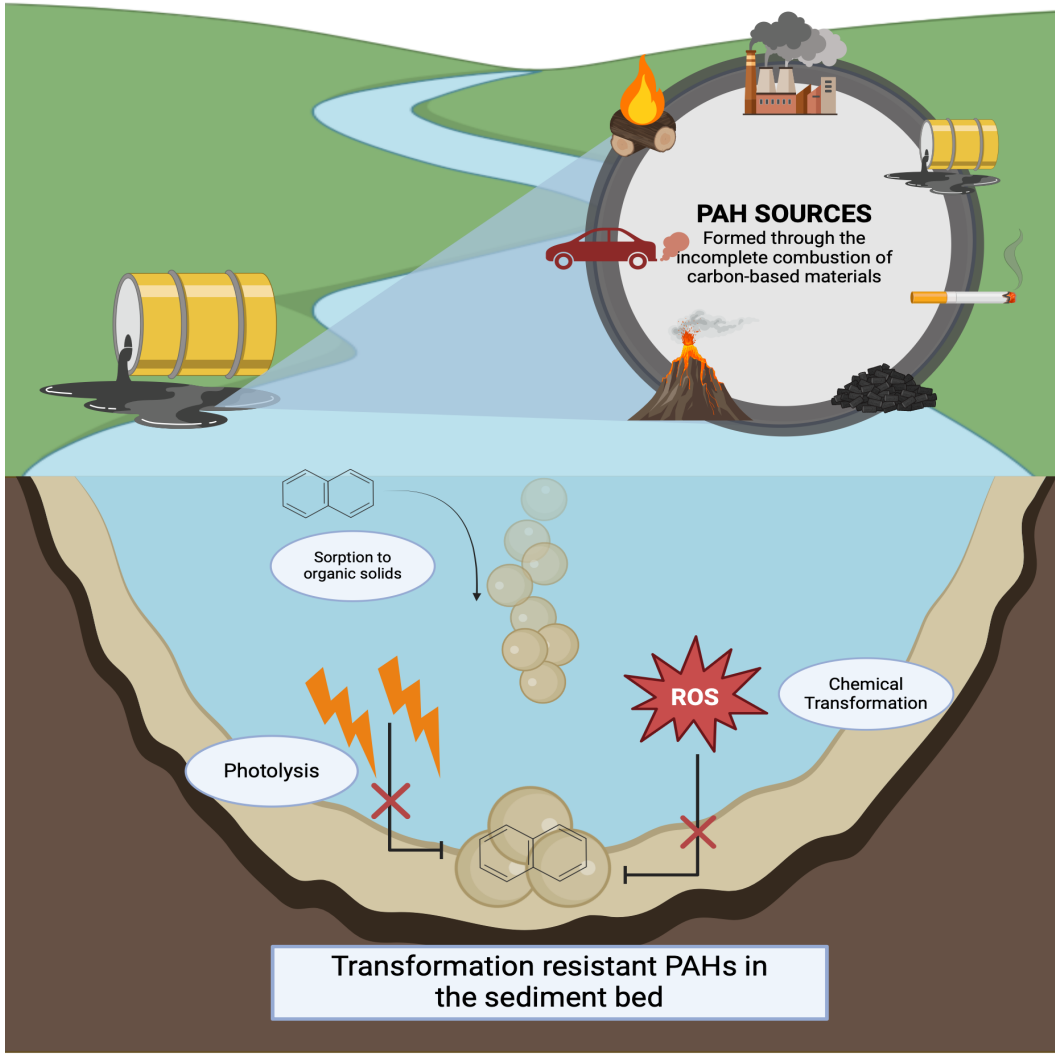
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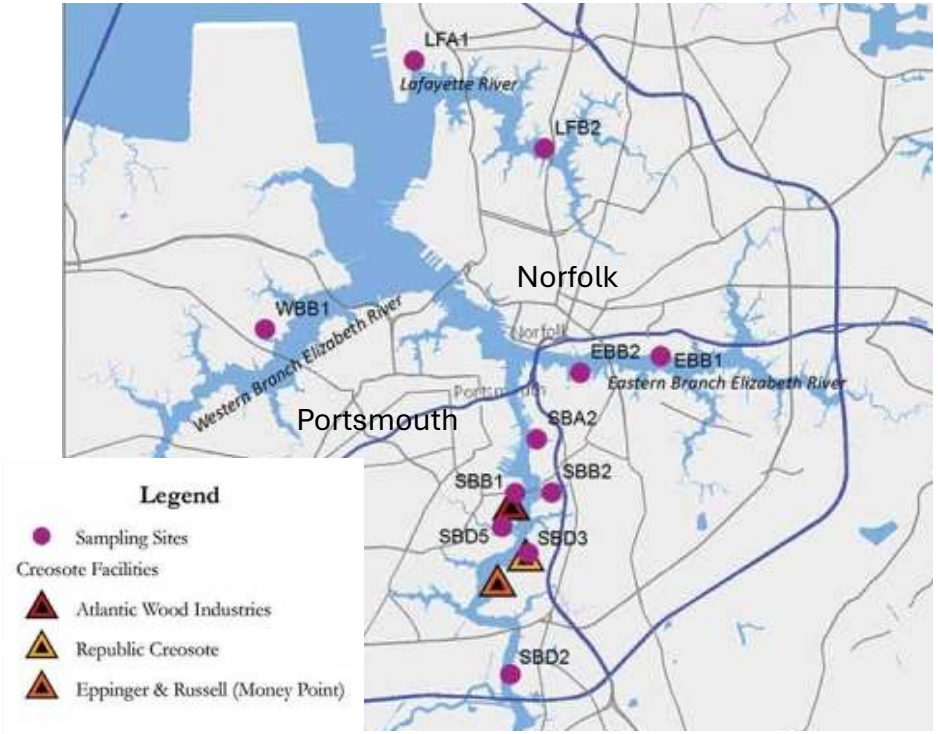
Chemistry Matters, 2021



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Elizabeth River Superfund Site

- Wood treatment facilities running through the mid-20th century spilled large amounts of creosote in the river
  - Creosote contains a broad range of PAHs
  - Hazardous levels of PAHs still observed 70 years later



Di Giulio & Clark, 2015



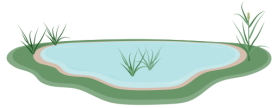
Library of Congress, 1943

Physical remediation approaches

Dredging removes the contaminated sediment bed and transports it to a secondary location



Usually a very **costly** process



Ecologically **invasive**, often causing additional issues



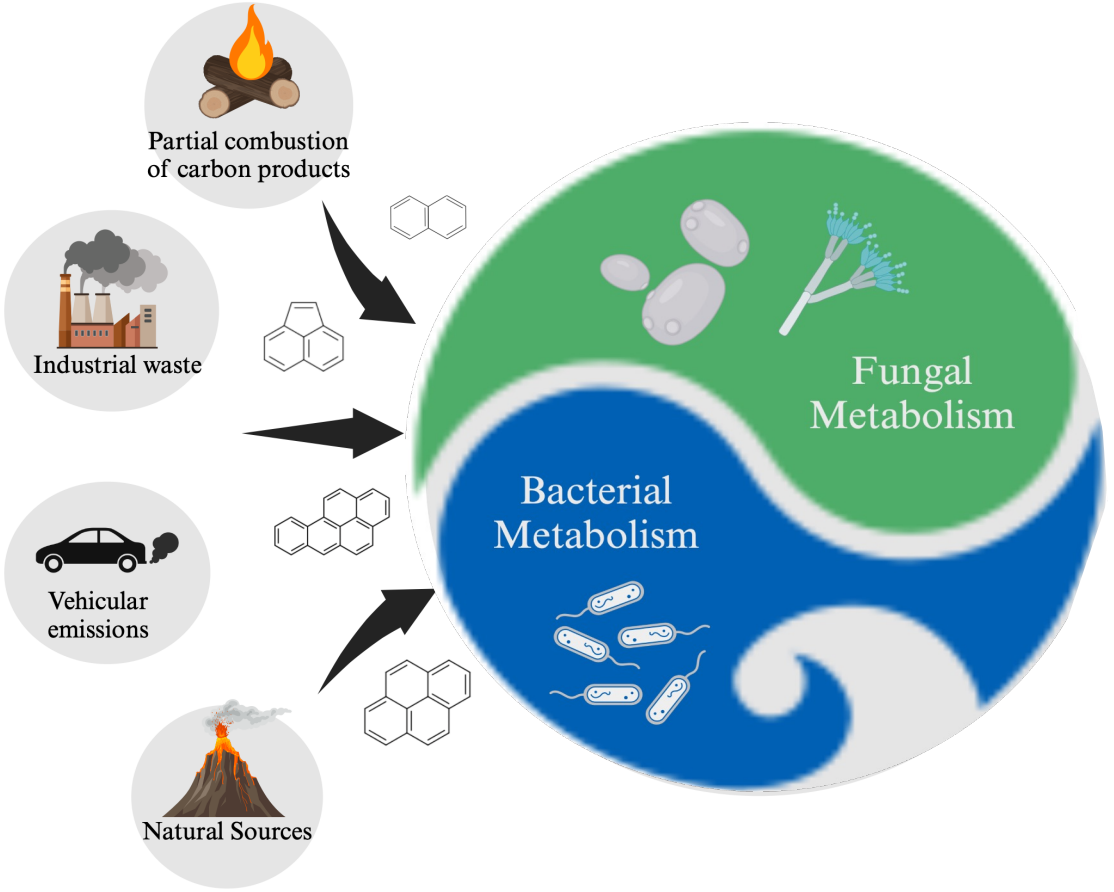
IADC, 2024

Who receives this untreated sediment?

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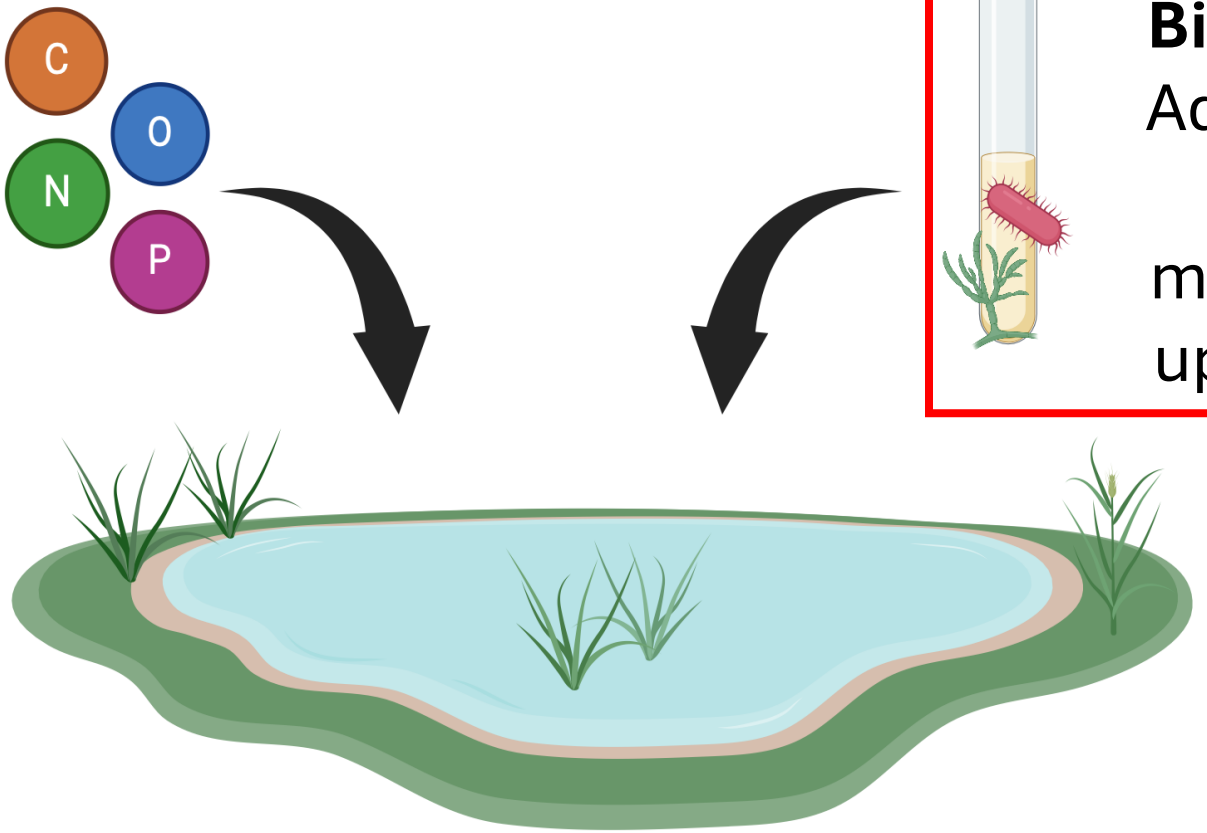
A sustainable, cost-effective alternative

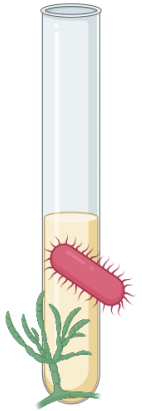
Bioremediation: Harnessing a microbe's metabolism to degrade contaminants into less toxic byproducts



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Biostimulation:
Stimulating the activity of native microbes through the addition of nutrients





Bioaugmentation:
Adding exogenous, PAH-degrading microbes to speed up bioremediation

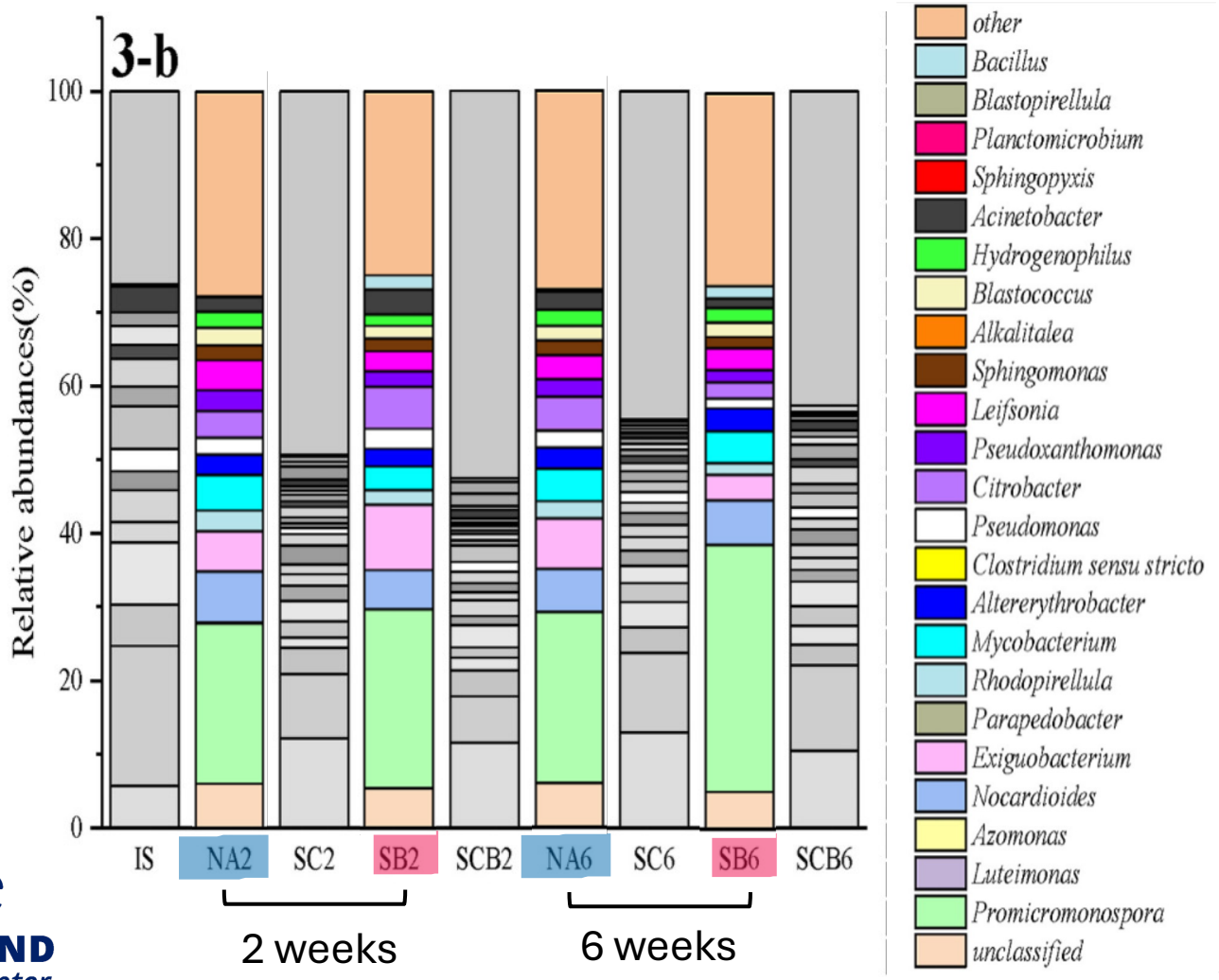


Natural attenuation:
Allowing the natural environment to degrade PAHs without human intervention

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Limitations of bioaugmentation

Exogenous strains struggle to establish in the environment



NA → Natural attenuation treatment

SB → Bioaugmentation treatment

Results suggest that bioaugmented strains don't compete well with indigenous microbes

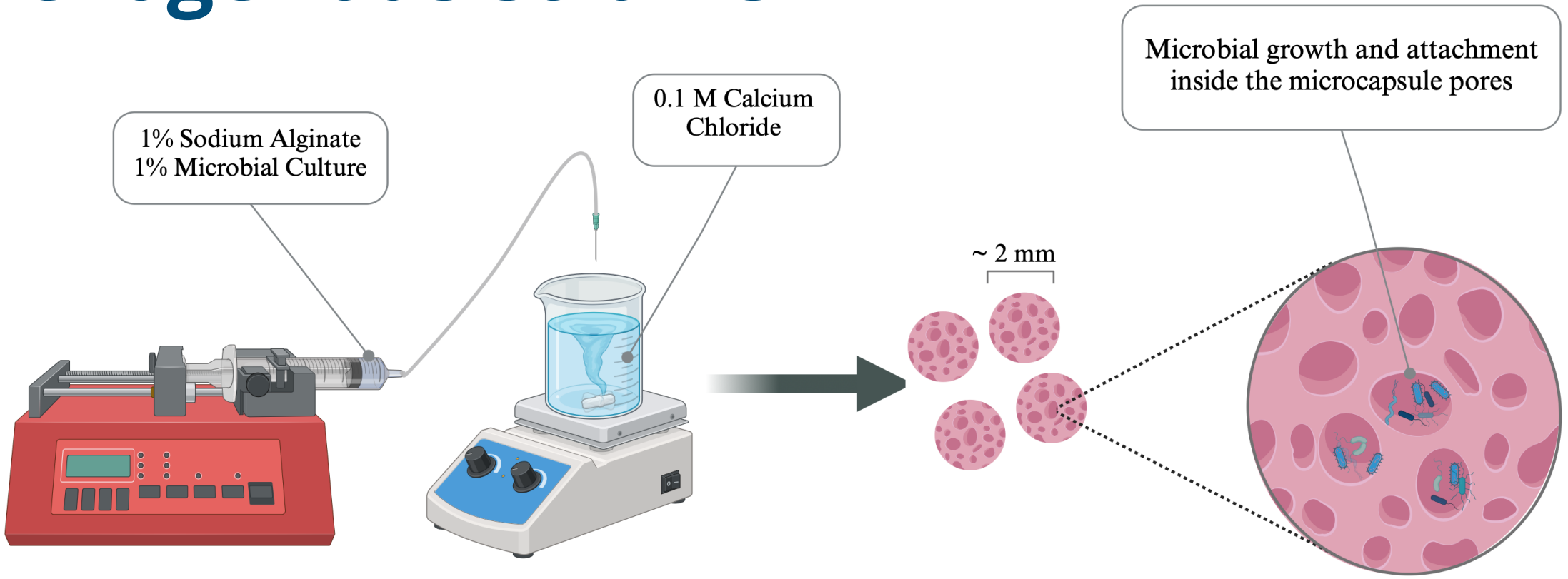
Wu et al, 2022

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The goal of this project

To develop innovative strategies that **enhance the growth and viability** of PAH-degrading microorganisms, improving the **efficiency of bioaugmentation**

Enhancing the establishment of exogenous strains



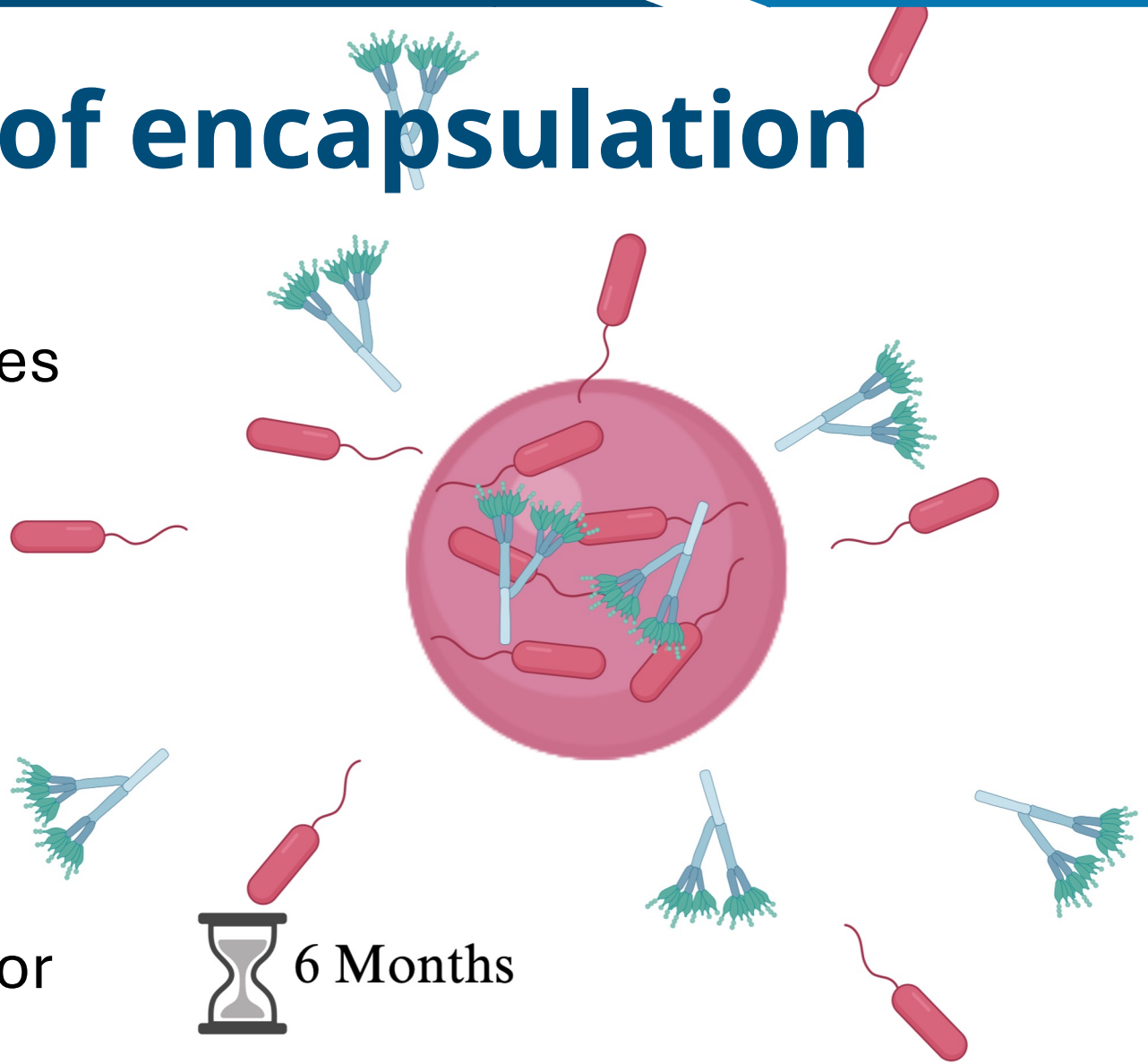
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Expected effects of encapsulation

Protective Barrier:
Encapsulation protects microbes from harsh environmental conditions

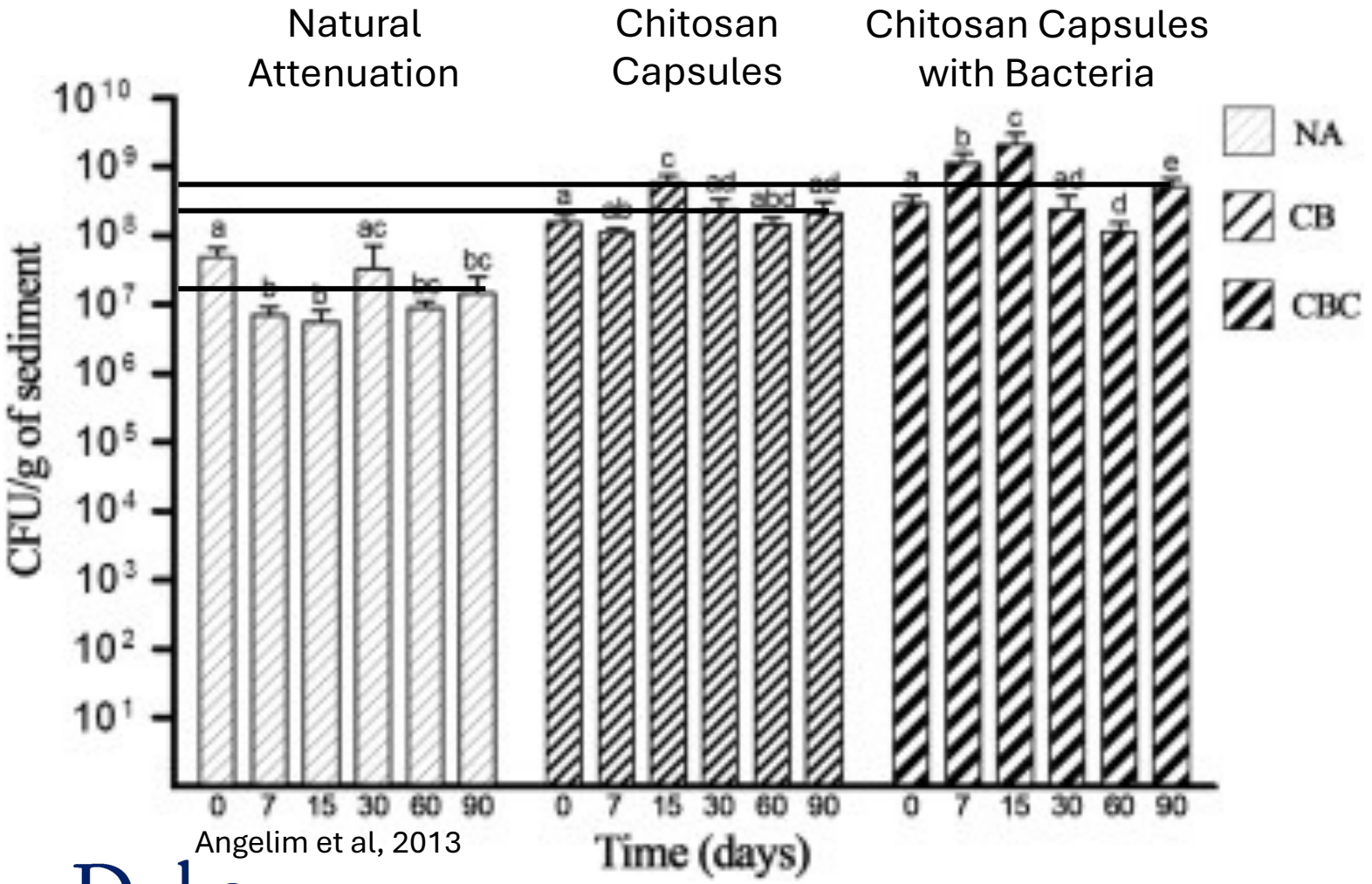
- pH
- Microbial competition
- Predation of other organisms

Continual Release:
Microcapsules allows for the continual release of microbes for stronger communities



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Encapsulation enhances establishment



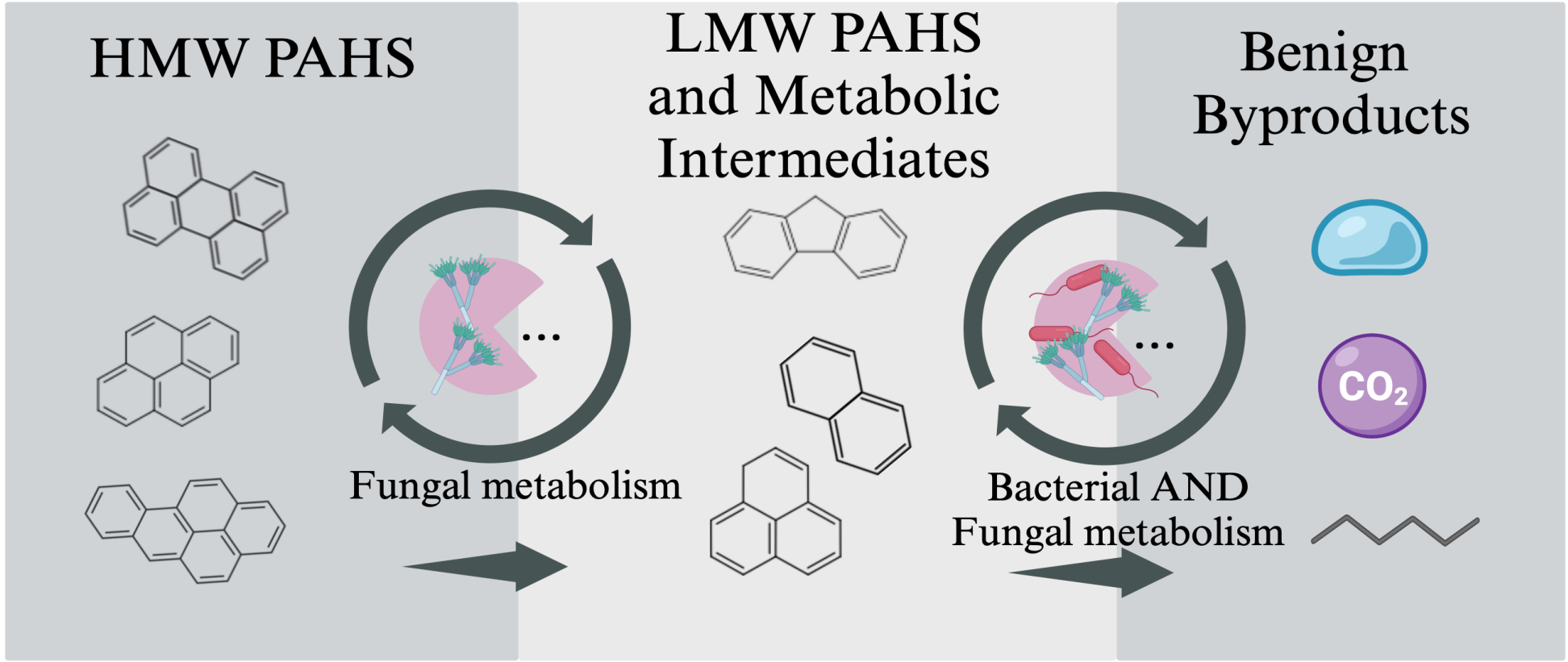
Angelim et al, 2013

Capsules enhance the establishment of microbes in sediment

- Shows an increase in CFU in mangrove sediments after 90 days

Capsules were placed in Mangrove sediments with bacteria present

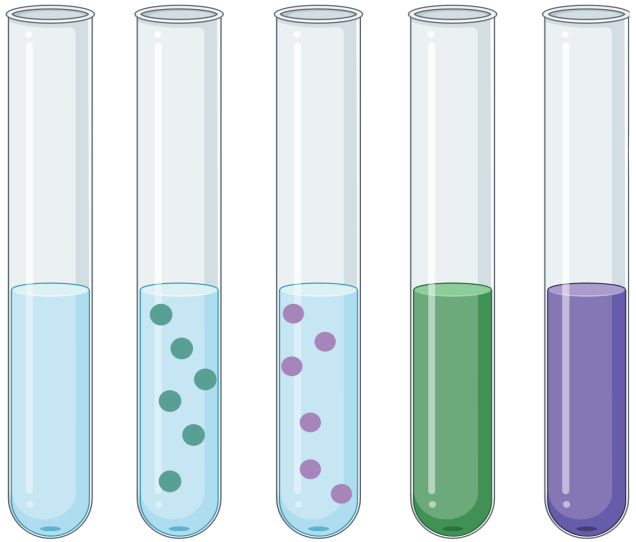
Interested in the synergistic relationships between bacteria and fungi



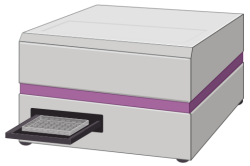
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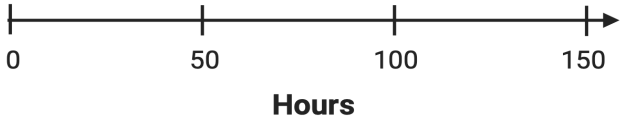
Methods to measure capsule effect

Did **growth, viability, and PAH degradation** measurements for both **bacteria and fungi**



encapsulated and free living cultures



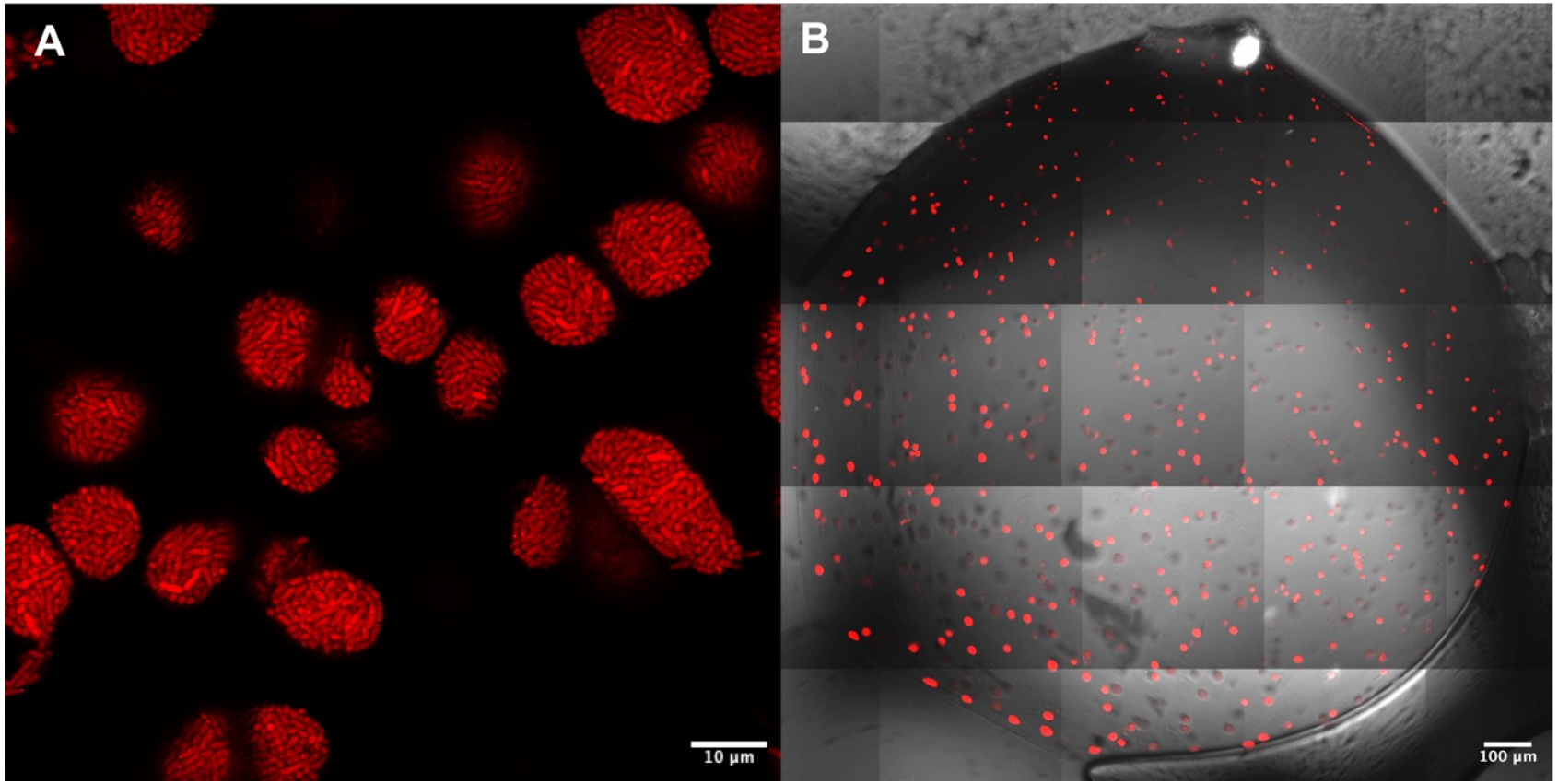


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Pseudomonas putida growth in capsules



Amelia Foley



Capsule maintains suitable habitat for survival and growth of bacteria

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Capsule formulation impacts bacterial viability after 8 weeks

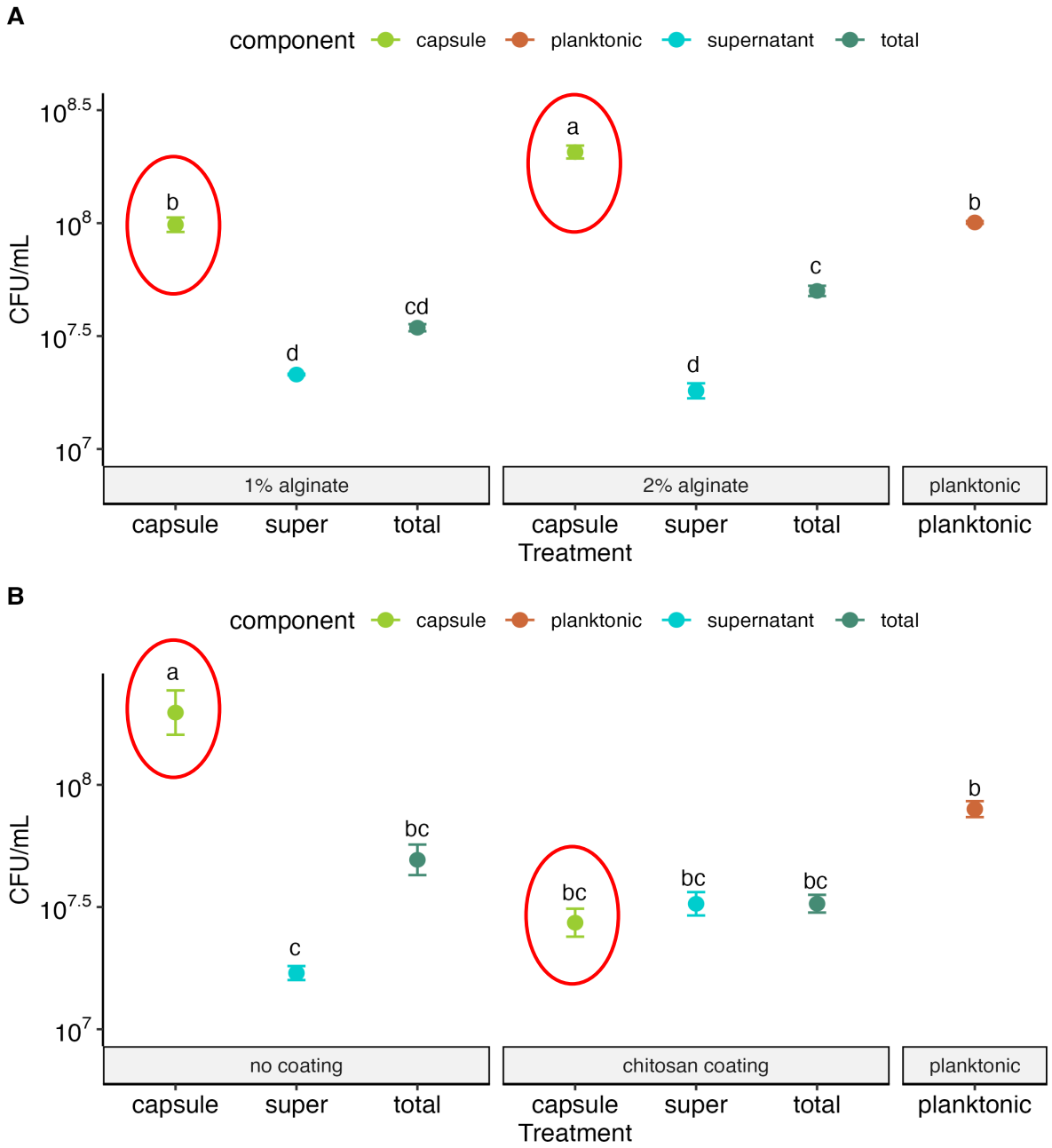


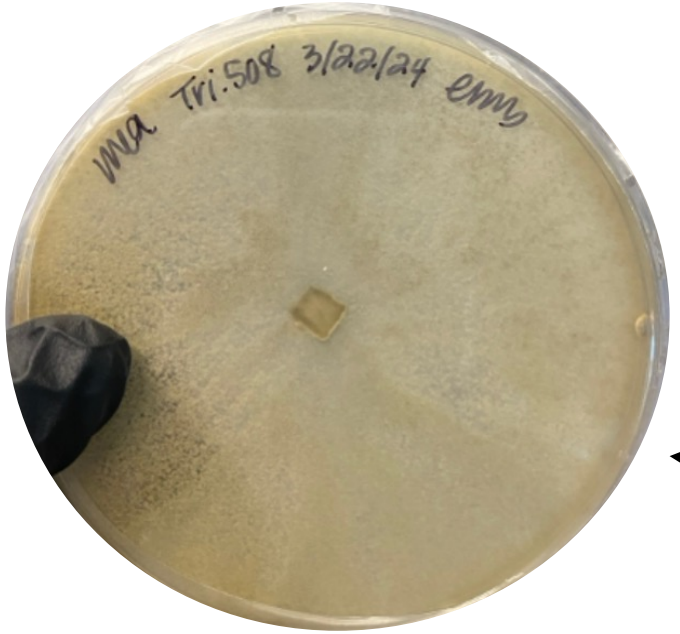
Figure 1. Viability/survival for *P. putida* G7 capsules formulated with varying A) alginate concentration and B) chitosan coating status. Error bars represent standard error (n = 3).

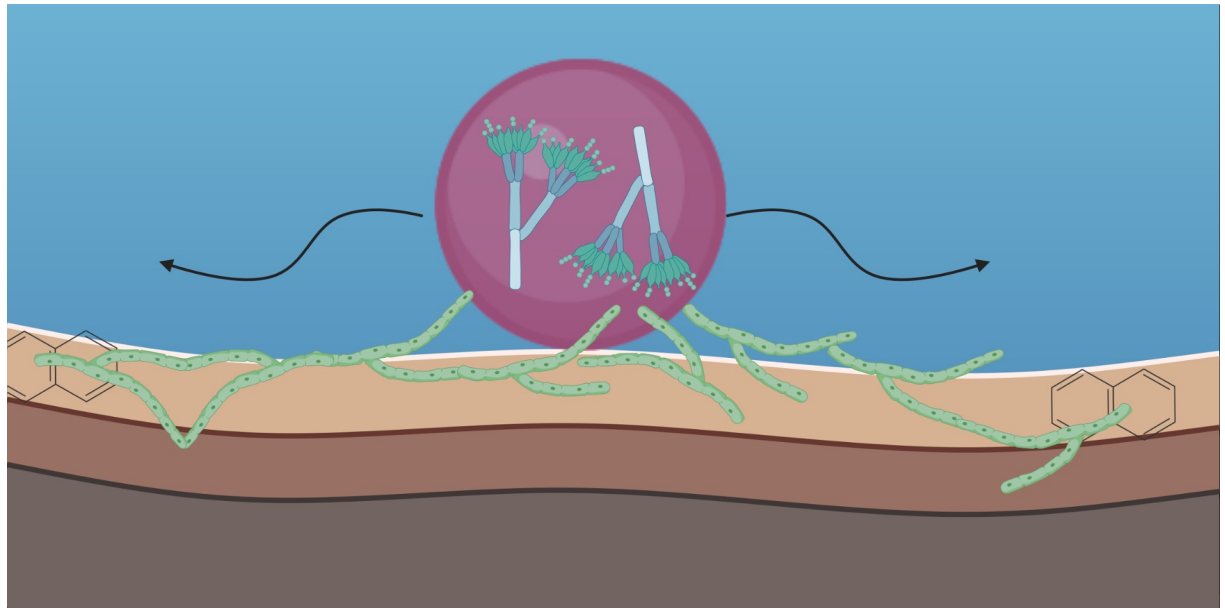
Higher capsule bacterial concentration in 2% alginate

Lower capsule bacterial concentration in chitosan-coated capsule

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Filamentous fungi *Trichoderma deliquescens*





extensive hyphae network

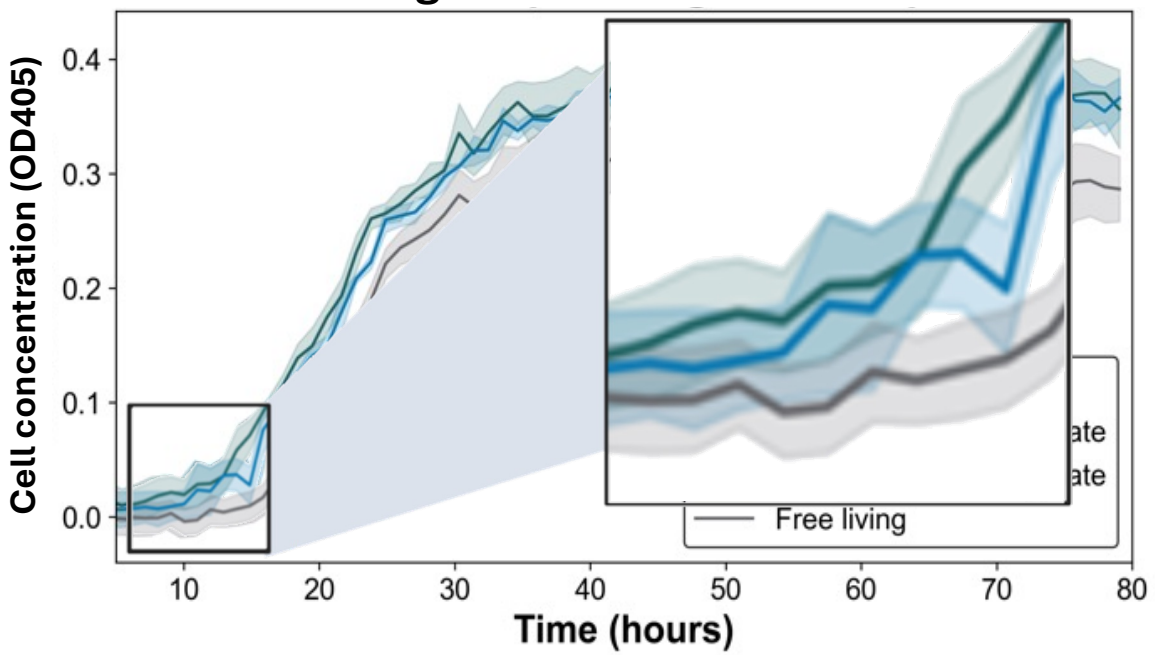
Bioaugmentation benefits of hyphal growth:

- Creates strong communities
- Can find PAHs more effectively
- Potential for O_2 transport

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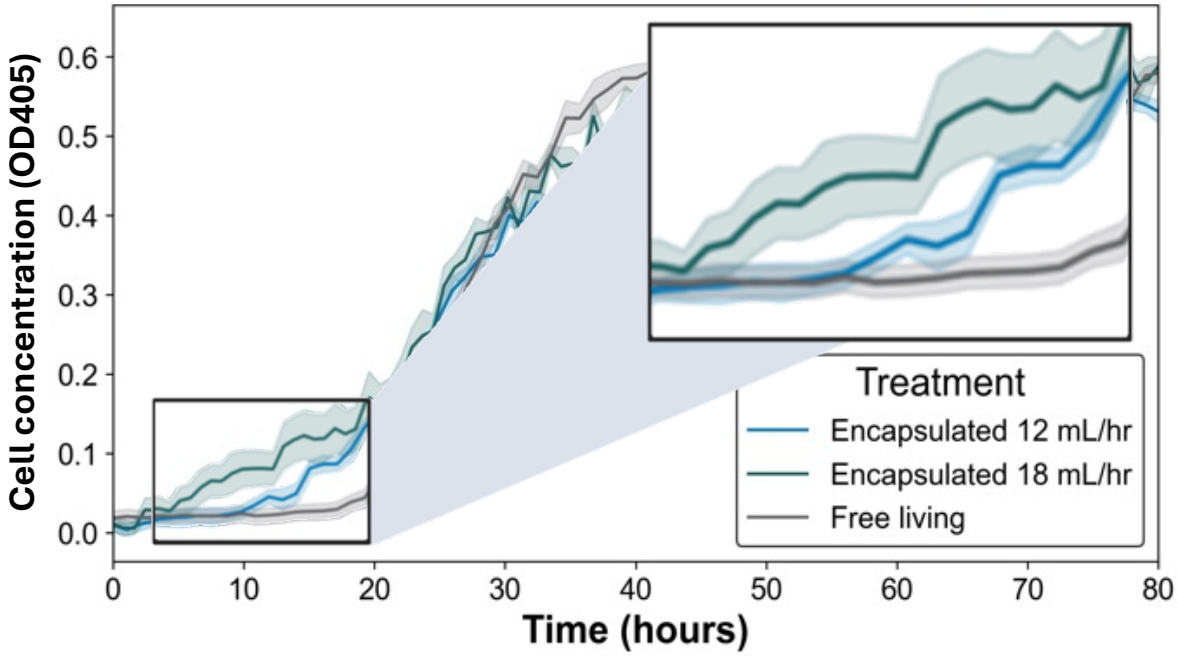
Encapsulation does not hinder growth of *Trichoderma deliquescens*

Alginate concentration



Change in capsule strength
2% more rigid

Extrusion rate

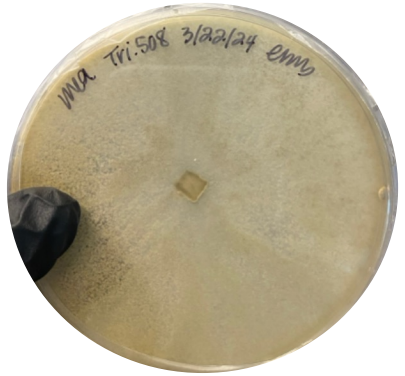


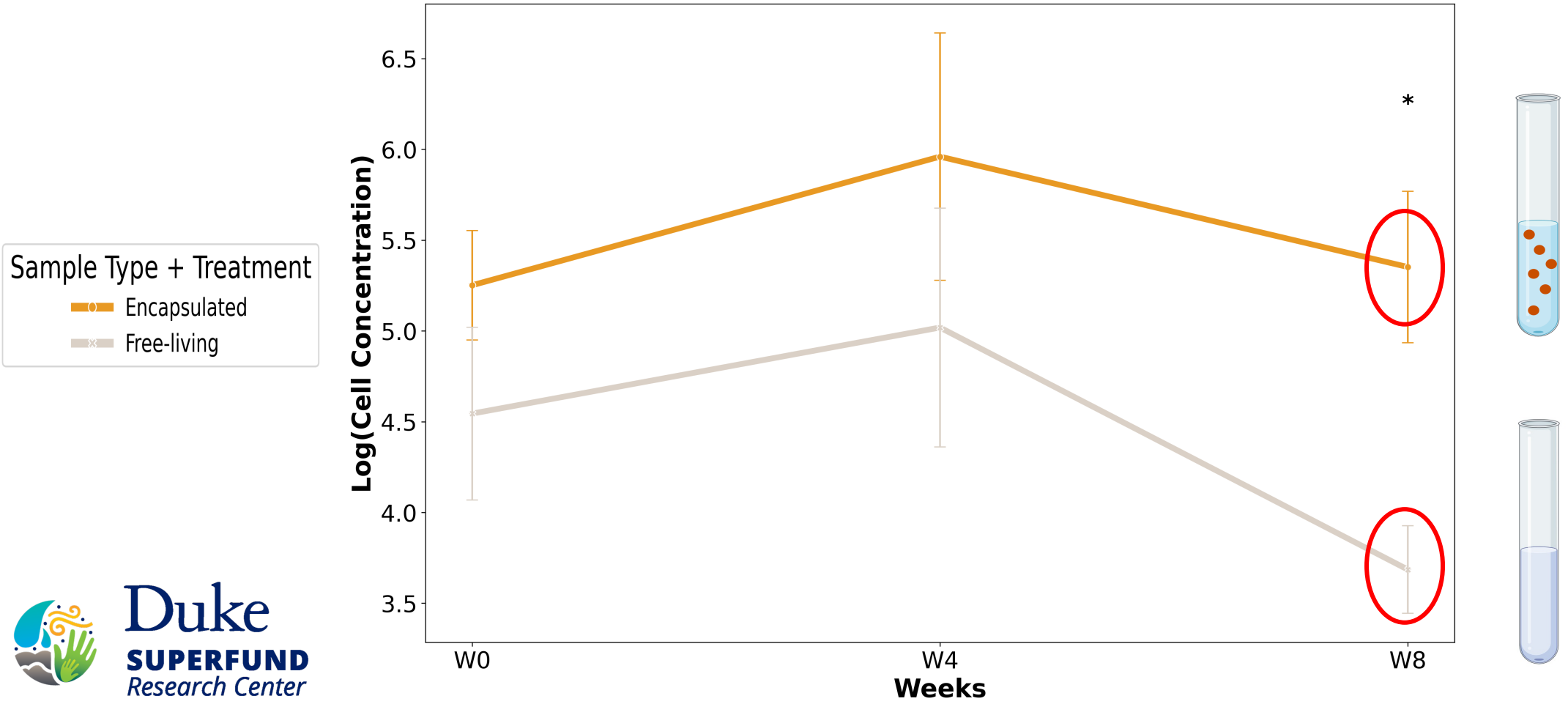
Change in capsule size
12 mL/hr larger

Capsules decrease the lag time for both capsule designs

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Encapsulated *Trichoderma* shows enhanced viability after 8 weeks

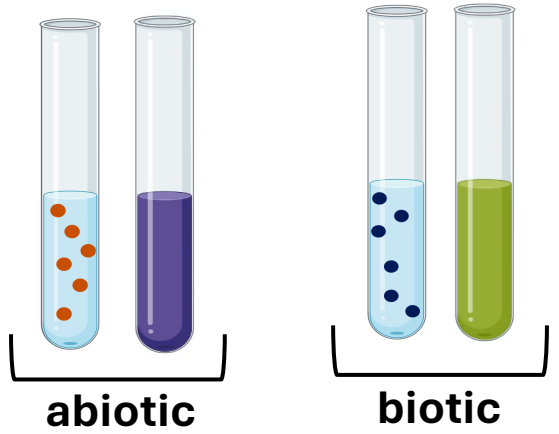


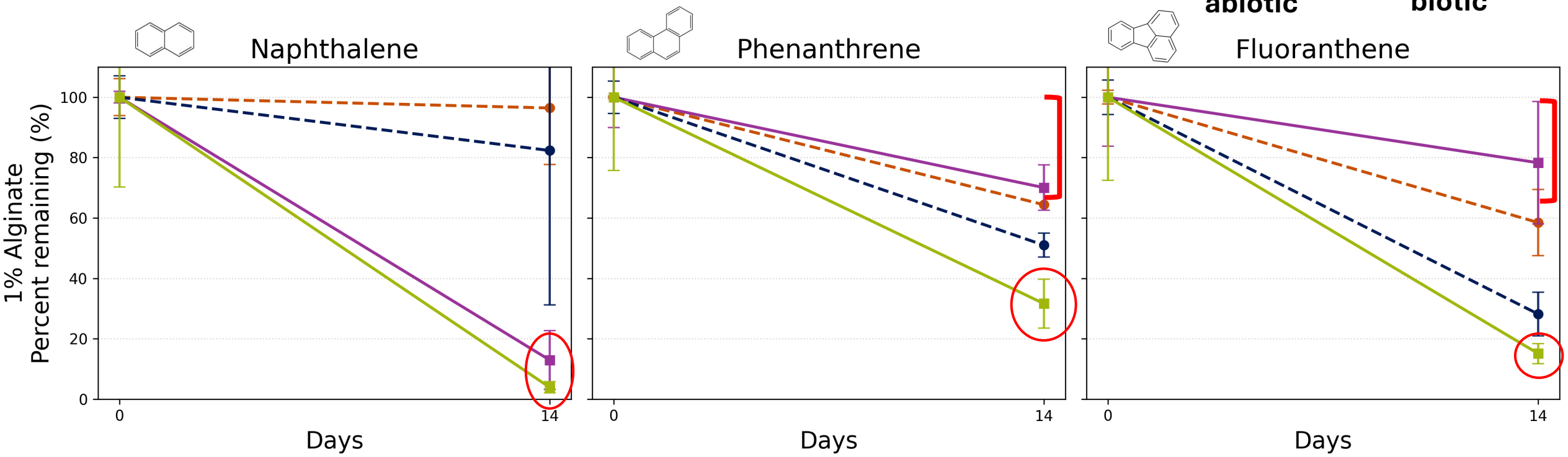


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Capsules affect PAH removal

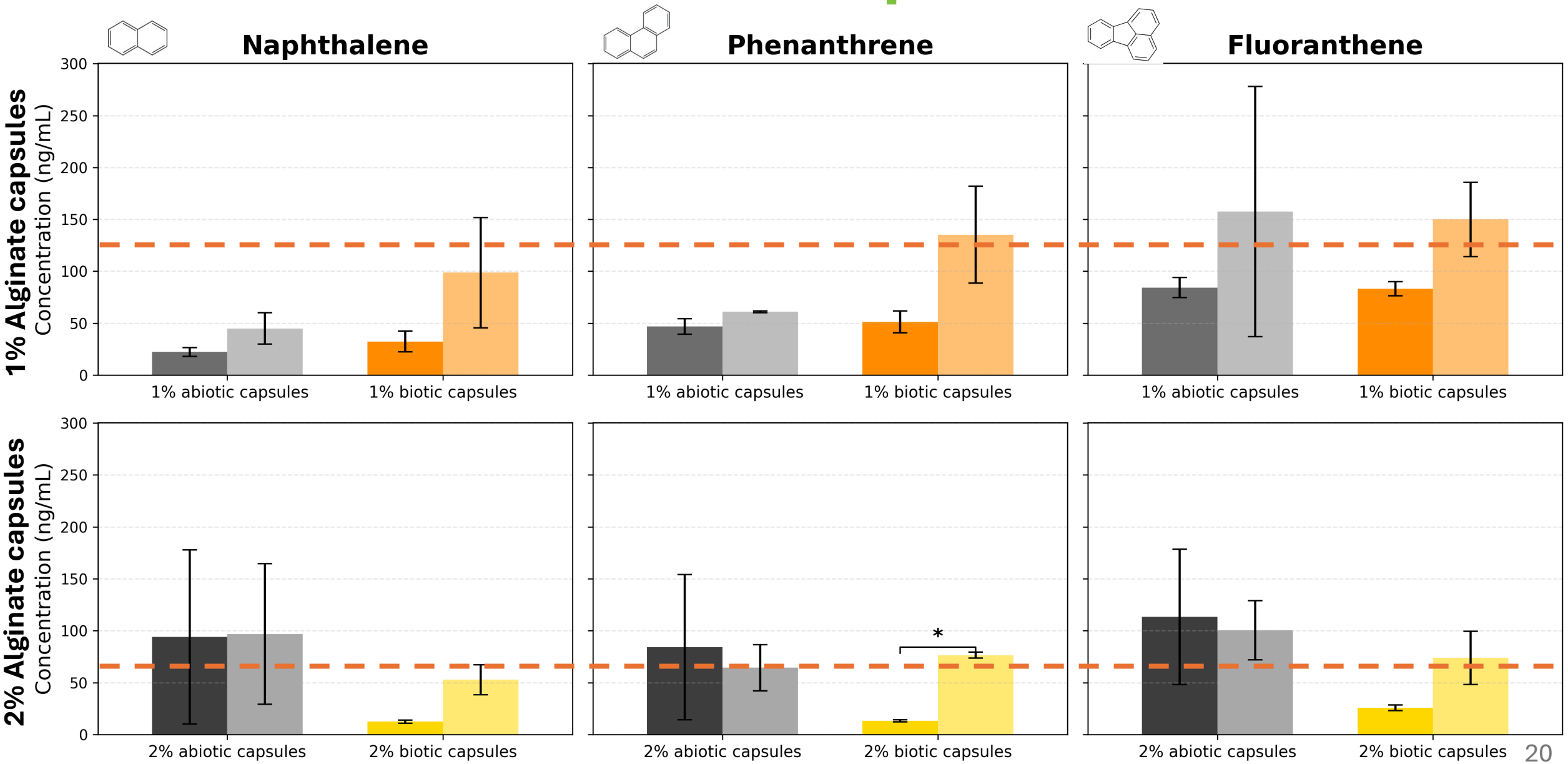
Capsules appear to accumulate PAHs, which may be advantageous for applications in the environment





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PAHs accumulate on capsules over 2 weeks



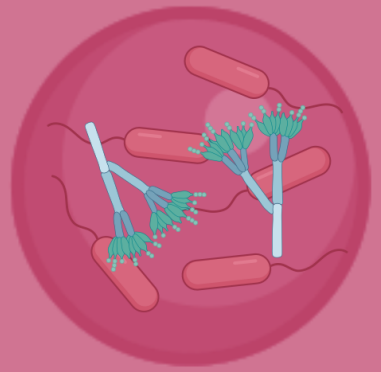
Conclusions

- Capsules can support bacterial and fungal growth
- Encapsulated cultures showed increased viability after 8 weeks
- Capsule formulation affects microbial success in simple systems
- PAHs sorb onto capsules and affect PAH removal in simple systems

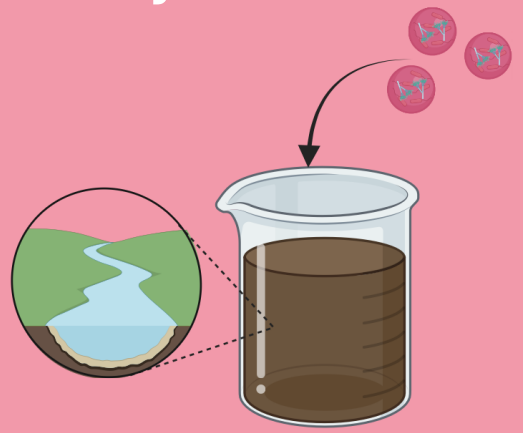
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Future experiments

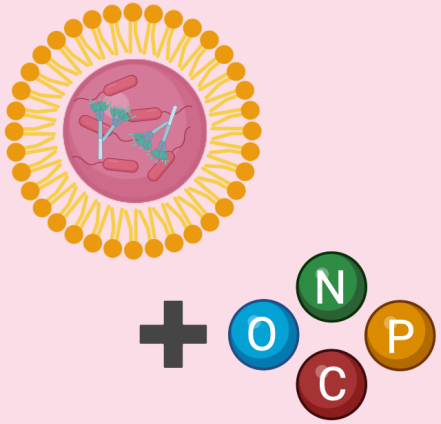
Adding consortia of bacteria and fungi to the capsules



How these capsules behave in more complex systems



Adding nutrients and chemicals to enhance growth and PAH degradation



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Acknowledgments

Project 5 Members

Dr. Claudia K. Gunsch

Dr. Heileen Hsu-Kim

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Amelia Foley

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Amelia Foley

William Stiffler

Qing Dai

Andrew Wrench

Kayla Fericy

Questions?